

SACS



Dynpac

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Dynpac program module generates dynamic characteristics including eigenvectors or natural mode shapes, eigenvalues or natural periods and modal internal load and stress vectors for a structure.

Because the Dynpac module provides the mode shapes and masses required for modal dynamic analysis, its execution is required prior to execution of any of the SACS dynamic programs.

1.2 PROGRAM FEATURES

Dynpac requires a SACS input model file or output structural data file and a Dynpac input file for execution. The program creates a common solution file containing normalized mode shapes, frequencies, internal loads etc. and a mass file.

Some of the main features and capabilities of Dynpac program module are:

1. Full six degree of freedom modes supported.
2. Guyan reduction of non-inertially loaded constrained degrees of freedom.
3. Generates structural mass and fluid added or virtual mass automatically.
4. Supports lumped or consistent mass generation.
5. User input lumped or consistent mass capability. User-defined joint weight in the Local Axis of the selected Member.
6. Ability to convert model input loading to mass.
7. Utilizes hydrodynamic properties and modeling from Seastate module.
8. Plate and beam element structural density overrides.
9. Member and member group fluid added mass property overrides.
10. Determines modal mass participation to allow determination of number of modes required for subsequent dynamic analyses.
11. Ability to override plate added mass coefficient.
12. Ability to override plate properties by plate group.
13. Includes P-Delta capabilities in addition to cable elements.

2 DYNAMIC MODELING AND INPUT

The Dynpac program requires a SACS model file or output structural data file and a Dynpac input file. The model file must contain minimal additional dynamic modeling information in order to perform the Dynpac analysis, namely, joint retained degrees of freedom (DOF) must be specified in the joint fixity columns on the appropriate 'JOINT' input line(s) and a 'LOAD' header must exist in the model file even if no loading is specified.

2.1 RETAINED DEGREES OF FREEDOM

Dynpac uses a set of retained degrees of freedom, selected by the user, to extract the Eigen values (periods) and Eigen vectors (mode shapes). All stiffness and mass properties associated with the constrained (reduced) degrees of freedom are included in the Eigen extraction procedure. The stiffness matrix is reduced to the retained degrees of freedom using standard matrix condensation methods. The mass matrix is reduced to the retained degrees of freedom using the Guyan reduction method assuming that the stiffness and mass are distributed similarly. All degrees of freedom which are non-inertial (no mass value) must be constrained degrees of freedom. After modes are extracted using the retained degrees of freedom, they are expanded to include full 6 degrees of freedom for all joints in the structure. The expanded modes are used for subsequent dynamic response analysis.

Any joint degree of freedom, X, Y and Z translation and/or rotation, to be retained for extraction purposes must be designated in the model. A joint DOF may be retained by specifying a '2' in the appropriate fixity column on the 'JOINT' input line. Specifying a '0' or leaving the fixity field blank designates the DOF as a constrained degree of freedom to be reduced. For example, to retain the X and Z translation degrees of freedom, specify '202' or '2 2' in columns 55-57 on the 'JOINT' line defining the joint.

Note: Columns 55, 56 and 57 pertain to global X, Y and Z translation respectively and columns 58, 59, and 60 to X, Y and Z rotation respectively.

Support degrees of freedom require no special modeling for dynamic purposes.

Note: Specifying a '2' or '0' for a particular DOF, has no effect for static analysis.

In dynamic analysis, to accurately calculate the effects of a concentrated mass along the length of a member it is best to include a joint at that location. Also, if a local mode due to the concentrated mass is important to the analysis, then the model should include retained degrees of freedom at the joint at the location of the mass. In this way, the dynamic analysis will use mass which is distributed in a manner that matches the mass distribution of the model.

2.2 STRUCTURAL MASS

2.2.1 Generating Structural Mass Automatically

By default, Dynpac generates structural mass for modeled beam, plate and shell elements automatically. Structural masses are also generated if 'SA' is specified as one of the execution options in columns 63-68 on the 'DYNOPT' line. Structural masses are not generated if option 'SO' is specified in columns 63-68.

Structural mass may be calculated as lumped or consistent mass by specifying 'LUMP' or 'CONS' in columns 15-18 on the 'DYNOPT' line respectively. The lumped method places all element mass at the nodes to which the element is connected while the consistent approach assumes mass is distributed along the element. Although, the default method is lumped, consistent mass may be desirable for structures immersed in fluid.

The following example indicates that the mass of modeled elements is to be calculated by the program in addition to converting some load cases in the model file to mass. The consistent mass approach is to be used.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
DYNOPT	CONS			SA-Z				

Note: Because the lumped approach does not generate mass moments of inertia, the weight moment of inertia for each rotational DOF retained must be specified in the Dynpac input file when using the lumped approach.

2.2.1.1 Default Structural Density

For a beam element, the density specified on the GRUP input line is used as the default when generating structural mass automatically, unless density is specified on the MEMBER line. If structural mass is not specified the density specified on the 'DYNOPT' line is used.

The density specified on the PGRUP or PLATE input lines located in the model file are used for plate elements. For shell elements on the other hand, the density specified in columns 19-25 on the DYNOPT line is used. The density specified on the 'SHELL' line is ignored by the Dynpac program module.

2.2.1.2 Overriding Structural Density

The density for individual members, plates, plate groups, shells and member groups may be overridden for mass generation purposes. The member, plate, shell or group name, along with the structural density override, are specified in the Dynpac input file on the MBOVR, PLOVR, PGOVR, SHOVR and GROVR override lines, respectively.

The following example specifies that the density of member 101-157, member group MM1, plate A101 and plate group PG1 is to be 100.0 for the purpose of determining the dynamic characteristics.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	MBOVR	101 157	100.0					
2	GROVR	MM1	100.0					
3	PLOVR	A101	100.0					
4	PGOVR	PG1	100.0					

2.2.1.3 Members Structural Torsional Mass

Dynpac can automatically calculate torsional mass of members to be included in model mass matrix. To enable this option, the user may enter 'MST' option on columns 74-76 of DYNOP2 input line. Dynpac uses consistent mass matrix formulation to compute torsional mass matrix in member local coordinates systems. The torsional mass matrix is given in local coordinates system as

$$\mathbf{M}_{torsion} = I_0 \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{3} \end{bmatrix}$$

where I_0 is polar moment of inertia of a given member with respect to its section centroid. I_0 can be determined based on section properties as

$$I_0 = \frac{m}{A} (I_{yy} + I_{zz})$$

in which m is total mass of the member, A is cross-section area and I_{yy} and I_{zz} are second moment of the area with respect to the local axes Y and Z, respectively.

Please consider following notes for torsional mass in Dynpac:

1. Consistent mass is required to include torsional mass and this option has no effect on lumped mass calculation.
2. This is only structural torsional mass – fluid mass is not considered in torsional mass calculation.
3. Tapered Member: A , I_{yy} and I_{zz} values are determined by averaging their values at the member two ends.
4. Multi-segmented Members: A , I_{yy} and I_{zz} values are computed through weighted averaging their values over all segments. The weight for a given segment is the ratio of segment length to member length.

2.2.2 Converting Loads to Mass Automatically

Loading contained in the SACS model file can be converted to structural joint or member mass automatically by specifying 'SA' as one of the execution options in columns 63-68 on the 'DYNOPT' input line.

The direction of loads to be converted and whether the same sign or the opposite sign of the load is to be used when converting to mass must also be specified in the execution options. If loading in the model file defined in the X direction is to be converted to mass, then '±X' should be specified. To convert loading defined in the Y or Z directions, '±Y' or '±Z' should be specified as one of the execution options respectively. The sign of the load direction specified, denotes whether the mass calculated from the load line will have the same sign as the load, designated by '+', or the opposite sign of the load designated by '-'. For example, when converting loading in the global -Z direction (such as gravity loading) to mass, the mass should have the opposite sign as the load specified (i.e. positive mass). Therefore, execution options 'SA-Z' (or 'SO-Z') should be specified on the 'DYNOPT' input line.

The following example indicates that the mass of modeled elements is to be calculated by the program in addition to converting load cases in the Z direction in the model file to mass. The sign of the mass will be the opposite of the sign of the load.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	JTWGT	601	10.0	10.0	10.0			
2	JTWGT	603	10.0	10.0	10.0			

Note: When converting loading to mass, the sign of the net load for any load vector must be such that no negative mass is introduced.

2.2.2.1 Designating Load Cases to Convert to Mass

When loads specified in the SACS model file or Seastate input file are to be converted to mass, only load cases specified on the LCSEL line(s) designated as dynamic load cases (i.e. function 'DY') are converted. For example, the following designates that load cases 4 and 5 are to be converted to mass by the program.

Note: Either the 'SA' or 'SO' options must be specified on the DYNOPT line in order to convert the designated load cases to mass.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LCSEL	DY		4	5			

Note: It is recommended to generate structural mass of the modeled structure automatically rather than converting the gravity loading created by Precede or Seastate.

2.2.2.2 Factoring Load Cases

Load Cases may be factored when converting to mass using the LCFAC line in the Seastate or model input file. In order to factor a load case, specify the load case and factor on the LCFAC using option 'DY'. For example, the following designates that 50% of load cases 4 and 5 are to be converted to mass.

Note: Load cases 4 and 5 are specified on the LCSEL and LCFAC lines.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LCSEL	DY		4	5			
2	LCFAC	DY	0.50	4	5			

2.2.3 User Input Joint Weight

Joint weights not defined in load cases designated to be converted to mass, may be specified as user defined concentrated joint weights in the Dynpac input file. Concentrated joint weights for X, Y and Z translational degrees of freedom and weight moments of inertia for the X, Y and Z rotational degrees of freedom are specified along with the joint name on the JTWGT line and are converted to masses automatically.

The joint weight can be also entered in a local axis of a member. The selected member should be defined in columns 72-75 and 77-80 of JTGWT line input. The local joint weight is only available for Consistent Mass calculation, any entered member will be ignored for Lumped Mass Matrix.

The following designates that X,Y and Z weight of 10.0 is to be applied at joints 601 and 603.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	JTWGT	601	10.0	10.0	10.0			
2	JTWGT	603	10.0	10.0	10.0			

2.2.4 Structural Mass Contingency Factors

Any mass generated by Dynpac or supplied as a load case in a SACS input file may be given a "contingency factor" via the 'DYNOP2' line. The contingency factor is a multiplier used to increase or

decrease the effect of the mass on structural loading. The contingency factor for structural mass generated by Dynpac is entered in columns 8-13; the contingency factor for masses entered as SACS load cases is entered in columns 14-19.

The 'DYNOPT' line in the following example specifies that loading in the -Z direction will be converted to structural mass. The 'DYNOP2' line specifies that Dynpac generated mass is to be given a contingency factor of 25% (1.25) whereas mass obtained from SACS loading in the -Z direction is to be given a contingency factor of 10% (1.10).

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DYNOPT	EN	CONS				SA-Z	
2	DYNOP2	1.25	1.10					

2.3 FLUID MASS

2.3.1 Generating Fluid Added Mass Automatically

For structures immersed in fluid, the added or virtual mass and the mass of entrapped fluid can be generated automatically. By default, the fluid mass, mudline elevation and the water depth are read from the model file or from the Seastate input data. If this data has not been previously specified in the model, it must be specified on the DYNOPT line (in the Dynpac input file) in columns 26-32, 33-39 and 40-46, respectively. The normal and axial added mass coefficients for members surrounded by fluid are input in columns 49-53 and 54-58 on the DYNOPT line.

Note: Values specified for fluid mass, mudline elevation and water depth will override any values input in the model file or in Seastate input data.

By default, the virtual mass is calculated based on the added mass coefficient in columns 49-53 on the DYNOPT line and actual member diameter unless an effective diameter is specified in columns 73-78 on the MEMBER input line. For plate elements, the virtual mass is determined using the added mass coefficient specified in columns 49-53 unless a value is indicated in columns 59-62 on the DYNOPT line.

The following specifies that the default added mass coefficient is 1.0 for beam elements and 0.01 for plate elements (i.e. effectively ignoring plate mass).

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DYNOPT	EN	CONS		1.0	0.01	SA-Z	

2.3.1.1 Member Overrides for Fluid Added Mass Generation

The effective member diameter used for added mass calculation may be overridden for individual members or for member groups using the 'MBOVR' or the 'GROVR' lines respectively in the Dynpac input file.

The following overrides the effective diameter of member 101-157 and member group MM1 to 0.001, thus ensuring that no added mass is calculated for these members.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	MBOVR	101 157	100.0					
2	GROVR	MM1	100.0					

2.3.1.2 Plate Overrides for Fluid Added Mass Generation

The added mass coefficient for plates and plate groups may be overridden using the PLOVR and PGOVR lines, respectively in the Dynpac input file. The following specifies that the plate added mass coefficient for plate A101 and plate group PG1 is 0.001.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	PLOVR A101		100.0					
2	PGOVR PG1		100.0					

2.3.2 Generating Fluid Entrapped Mass Automatically

Entrapped mass is calculated for members designated as flooded in the model file based on the actual diameter of the member.

2.3.2.1 Member Overrides for Fluid Entrapped Mass Generation

The flood condition may be overridden for all members on the DYNOPT line in columns 47-48. The flood condition for individual members or member groups may be changed using the MBOVR or the GROVR line images in the Dynpac input file.

The following overrides the flood condition of member 101-157 and member group MM1 to non-flooded, thus ensuring that no entrapped mass is calculated for these members.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	MBOVR N 101 157		100.0					
2	GROVR MM1 N		100.0					

Note: The flood condition specified on the 'DYNOPT' line overrides any existing flood condition for all members in the model unless flood condition is changed with subsequent 'MBOVR' or 'GROVR' lines.

2.4 HYDRODYNAMIC MODELING USING SEASTATE

The Seastate program can be used to account for the hydrodynamic effects of unmodeled structural items and/or marine growth as well as the weight of grout in the annulus of concentric tubular sections (assuming 150 lb/ft³ grout density). Seastate updates the member lines to account for the density and effective diameter due to marine growth specified on 'MGROV' lines in the SACS model or in the Seastate input file. Member density is also updated to reflect the effective density based on any density and/or cross section area overrides specified in the Seastate input. The effective member diameter in columns 73-78 on the 'MEMBER' input line is updated to account for any local Y and Z force dimension overrides specified (in addition to effects of marine growth).

Note: Seastate must be executed with 'DYN' specified in columns 56-58 on the 'LDOPT' line in the Seastate input file or with the appropriate option specified in the Executive in order to generate hydrodynamic properties. The model updates are contained in the output structural data file created. See the Seastate User's Manual for a detailed discussion.

2.5 SIMULATING NON-LINEAR FOUNDATIONS

Because the dynamic capabilities in the SACS system use linear theory (i.e. modal superposition), non-linear foundations must be represented with a linearly equivalent system. The equivalent linear foundation model must be incorporated into the SACS model for the purposes of dynamic analysis.

Note: The Pile program module can be used to determine the length, properties and offsets for equivalent pile stub elements used to represent the soil-pile interaction. See the PSI/Pile program user's manual for a detailed discussion.

2.5.1 Including Linearized Foundation Automatically

The PSI program may be used to generate an equivalent foundation stiffness matrix or super-element to be used to represent the foundation for dynamic analysis. The equivalent foundation super-element may be included as part of the model by specifying 'I' in column 9 of the OPTIONS line in the model file or by selecting the appropriate superelement option in the Executive.

2.6 INCLUDING P-DELTA EFFECTS

The Dynpac program can include the effects of P-Delta on the dynamic characteristics of the structure. This feature allows the user to designate reference load case(s) representing static dead loading on the structure.

To include P-delta effects, the p-delta effects option must be designated in the model file using the OPTIONS line. The reference load cases must then be designated in the model file or the Seastate input file using the LCSEL line with the 'PD' option. For example, the following shows that dead loading defined by load cases DEAD, EQPT and AREA are to be used to determine the P-delta effects on the beam elements.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	OPTIONS	EN PD	1	DC	C			
2	LCSEL PD	DEAD EQPT AREA						

Load factors may be applied to the reference load cases using the LCFAC line. For example, in the following, 50% of load cases DEAD EQPT and AREA are used to obtain the reference axial load.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	LCSEL PD	DEAD EQPT AREA						
2	LCFAC PD	0.5	DEAD EQPT AREA					

Note: Dead loads are typically used as P-Delta loads. For cable elements, the pre-tension load should be designated as the P-Delta load.

3 DYNPAC TROUBLE SHOOTING

3.1 MODEL STIFFNESS MATRIX

As part of the dynamic characteristic analysis, the Solve module is used to generate the stiffness matrix properties of the structure. The structural model matrix created by Solve must be 'Positive Definite' in order to determine the dynamic characteristics of the structure. In general, if a degree of freedom for any joint or portion of the structure is not restrained by fixity or by stiffness from other elements, the matrix will be 'Non'-Positive Definite'. For further discussion on matrix 'Non-Positive Definite', see the section titled 'SACS IV Trouble Shooting' in the SACS IV user's manual.

The Solve module also determines the accuracy of the solution and reports it as the 'Maximum Number of Significant Digits Lost'. In general, solutions with six or fewer significant digits lost are sufficiently accurate while solutions with twelve or more lost are not. The SACS IV user's manual addresses possible causes for excessive numbers of lost significant digits.

3.2 MODEL MASS MATRIX

The structural mass matrix is developed by the Dynpac program module. Like the stiffness matrix, the structural mass matrix must be 'Positive Definite' in order for it to be inverted. When the mass matrix cannot be inverted, the message 'Non-Positive Definite Mass Matrix' is printed in the listing file. Some common reasons for the structural mass matrix becoming 'Non-Positive Definite' are as follows:

1. No degrees of freedom in the model are retained DOFs. The error message will normally refer to a degree of freedom for joint name 0.
2. All degrees of freedom are retained DOFs so that there are no constrained DOFs.
3. The mass for a particular degree of freedom is negative. This can occur when converting loads specified in the model file to mass using the 'SA' or 'SO' option on the DYNOPT line. When negative loads in the model file are to be converted, i.e. gravity loads, the '-X', '-Y' or '-Z' option should be specified so that the sign of the mass generated will be positive (opposite to that of the load).
4. A rotational degree of freedom is a retained DOF but no mass moment of inertia was generated (i.e. lumped approach) or no weight moment of inertia was specified in the input file for that DOF.

When a matrix 'Non-Positive Definite' occurs, the critical degree of freedom and the joint name are reported in the Dynpac listing file. For additional information on debugging the model, see the SACS IV user's manual.

4 COMMENTARY

4.1 STIFFNESS MATRIX REDUCTION

The purpose of the Dynpac program module is to generate dynamic characteristics (mode shapes and frequencies) of a structure. A Guyan reduction is performed to reduce the structural stiffness matrix K created by SACS IV as follows:

$$K = \begin{bmatrix} K_{rr} & K_{rc} \\ K_{cr} & K_{cc} \end{bmatrix}$$

where the subscript r designates retained degrees of freedom and the subscript c designates constrained degrees of freedom. Knowing that $F = K\delta$ or

$$\begin{Bmatrix} F_r \\ F_c \end{Bmatrix} = \begin{bmatrix} K_{rr} & K_{rc} \\ K_{cr} & K_{cc} \end{bmatrix} \begin{Bmatrix} \delta_r \\ \delta_c \end{Bmatrix}$$

the following relationships can be made.

$$F_r = K_{rr}\delta_r + K_{rc}\delta_c \quad (1)$$

$$F_c = K_{cr}\delta_r + K_{cc}\delta_c$$

If, by definition, no external forces are applied directly to the constrained degrees of freedom such that $F_c=0$, δ_c can be expressed as follows:

$$\delta_c = -\frac{K_{cr}}{K_{cc}}\delta_r \quad (2)$$

Substituting for δ_c in equation (1) yields a relation that can be used to calculate the external forces on retained degrees of freedom, namely,

$$F_r = K_{rr}\delta_r + K_{rc}\left(-\frac{K_{cr}}{K_{cc}}\delta_r\right) = \left(K_{rr} - K_{rc}\frac{K_{cr}}{K_{cc}}\right)\delta_r \quad (3)$$

Or

$$F_r = [K'_{rr}]\delta_r \quad (4)$$

where K'_{rr} is the reduced stiffness matrix. Once the retained degrees of freedom are calculated, relation (2) may be used to determine the constrained degrees of freedom.

4.2 MASS MATRIX GENERATION

4.2.1 Consistent Mass Approach

The mass matrix may be generated based on the lumped or consistent mass approach. The consistent mass generation approach represents the kinetic energy of the distorted element by the element joint velocities, as represented by the velocities of all degrees of freedom at the joint. The deflection δ and velocity $\dot{\delta}$ along a member may be expressed as follows:

$$\delta = f(x, \delta_a, \theta_a, \delta_b, \theta_b) \quad \dot{\delta} = f(x, \dot{\delta}_a, \dot{\theta}_a, \dot{\delta}_b, \dot{\theta}_b)$$

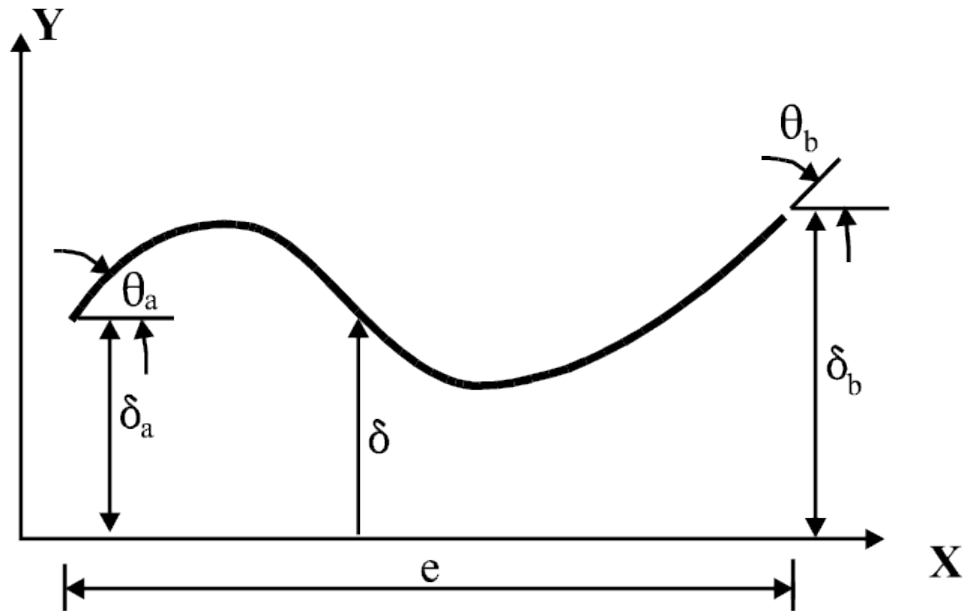


Figure 1. Beam element deflection model

The kinetic energy is defined as:

$$KE = \frac{1}{2} \int M \dot{\delta}^2 dx$$

where M is the mass per unit length. Taking

$$\frac{d}{dt} \left(\frac{dKE}{d\dot{q}} \right) \quad \text{for} \quad \dot{q} = \dot{\delta}_a, \dot{\delta}_b, \dot{\theta}_a, \dot{\theta}_b$$

results in

$$\begin{bmatrix} M \end{bmatrix} \begin{Bmatrix} \ddot{\delta}_a \\ \ddot{\theta}_a \\ \ddot{\delta}_b \\ \ddot{\theta}_b \end{Bmatrix}$$

where [M] is the elemental mass matrix for the element. The elemental mass matrix is then transformed into the global coordinate system and added to the overall structural mass matrix.

Note: Because the consistent approach takes into account the distribution of mass along the element, the mass matrix created includes off-diagonal coupling terms between all degrees of freedom, including rotational DOFs.

4.2.2 Lumped Mass Approach

In the lumped approach, a diagonal mass matrix is created by dividing each element mass into equal components along the global X, Y and Z directions and concentrating these masses at the end joints. Rotational mass or mass moments of inertia are neglected along with any off diagonal terms of the mass matrix.

Note: Because off diagonal terms are assumed to be zero in the lumped mass approach, it is not recommended when the element mass is not the same in all three directions such as when including effects of fluid added or virtual mass acting normal but not tangential to the element.

4.3 MASS MATRIX REDUCTION

After the overall mass matrix has been generated by either the consistent or lumped mass approach, it is partitioned into the same form as the stiffness matrix such that:

$$KE = \frac{1}{2} \left\{ \begin{matrix} \dot{\delta}'_m & \dot{\delta}'_s \end{matrix} \right\} \begin{bmatrix} M_{mm} & M_{ms} \\ M_{sm} & M_{ss} \end{bmatrix} \left\{ \begin{matrix} \dot{\delta}_m \\ \dot{\delta}_s \end{matrix} \right\}$$

Note: The terms M_{ms} and $M_{sm} = 0$ and M_{mm} and M_{ss} are diagonal matrices for the lumped approach.

Differentiating equation (2) with respect to time yields,

$$\dot{\delta}_s = -K_{ss}^{-1} K_{sm} \dot{\delta}_m$$

therefore, the equation for kinetic energy becomes

$$KE = \frac{1}{2} \left\{ \dot{\delta}'_m \right\} \begin{bmatrix} I & -K_{ms} K_{ss}^{-1} \end{bmatrix} \begin{bmatrix} M_{mm} & M_{ms} \\ M_{sm} & M_{ss} \end{bmatrix} \begin{bmatrix} I \\ -K_{ss}^{-1} K_{sm} \end{bmatrix} \left\{ \dot{\delta}_m \right\}$$

which is a standard Guyan reduction resulting in

$$KE = \frac{1}{2} \left\{ \dot{\delta}'_m \right\} [M'_{mm}] \left\{ \dot{\delta}_m \right\}$$

where M'_{mm} is the reduced mass matrix.

4.4 CALCULATING RESULTS

Once the reduced stiffness and reduced mass matrices are generated, the eigenvalues/eigenvectors for the retained degrees of freedom are extracted using the standard Householder-Givens extraction technique. The resulting eigenvectors at the retained degrees of freedom are expanded to obtain results for the reduced or constrained degrees of freedom which allows the calculation of modal reactions and modal elemental internal loads.

4.5 FLUID ADDED OR VIRTUAL MASS

Morrison's equation is used to determine the hydrodynamic loading due to fluid added or virtual mass. The resultant force per unit length, F , has a component normal to the element, F_n , and a component tangential or along the cylinder axis, F_t .

$$\bar{F} = \bar{F}_n + \bar{F}_t$$

where F_n and F_t are functions of the fluid relative velocity V_{rel} , fluid acceleration V' and the acceleration of the structure V'_s , and are given by the following for tubular elements:

$$\begin{aligned}\bar{F}_n &= \frac{1}{2} C_{Dn} D \rho_s \bar{V}_{rel_n} |\bar{V}_{rel_n}| + \frac{1}{4} \pi C_{Mn} D^2 \rho_s \bar{V}_n + (C_{Mn} - 1) \frac{\pi D^2}{4} \rho \bar{V}_{s_n} \\ \bar{F}_t &= \frac{1}{2} C_{Dt} D \rho_s \bar{V}_{rel_t} |\bar{V}_{rel_t}| + \frac{1}{4} \pi C_{Mt} D^2 \rho_s \bar{V}_t + (C_{Mt} - 1) \frac{\pi D^2}{4} \rho \bar{V}_{s_t}\end{aligned}$$

where the term $(C_m - 1)(\pi D^2/4)\rho$ is the fluid added mass term. The normal added mass, m_n , and axial or tangential added mass, m_t , may be rewritten as follows:

$$\begin{aligned}m_n &= (C_{Mn} - 1) \frac{\pi D^2}{4} \rho = C_{v_n} \frac{\pi D^2}{4} \rho \\ m_t &= (C_{Mt} - 1) \frac{\pi D^2}{4} \rho = C_{v_t} \frac{\pi D^2}{4} \rho\end{aligned}$$

where C_{v_n} and C_{v_t} are the normal and axial added mass coefficients input into the Dynpac program, respectively.

Note: Because the default tangential mass coefficient, C_{vt} , is zero, tangential added mass is ignored by default unless the coefficient is overridden by the user.

The added mass normal to the member and the mass tangential, if applicable, are broken into global X, Y and Z direction masses then added to the elemental mass matrix. Including the hydrodynamic inertial terms due to structural acceleration in the mass matrix, results in the automatic inclusion of acceleration dependent hydrodynamic forces including relative acceleration effects.

The global X, Y and Z components, m_{nx} , m_{ny} and m_{nz} , of the normal fluid added or virtual mass and the X, Y and Z components of the tangential fluid added mass, m_{tx} , m_{ty} and m_{tz} , are taken as:

$$\begin{aligned}m_{nx} &= m_n \cos \alpha_x & m_{ny} &= m_n \cos \alpha_y & m_{nz} &= m_n \cos \alpha_z \\ m_{tx} &= m_t \sin \alpha_x & m_{ty} &= m_t \sin \alpha_y & m_{tz} &= m_t \sin \alpha_z\end{aligned}$$

where α_x , α_y and α_z are the angle between the plane normal to the element and the global X, Y and Z axes respectively. See the following figure.

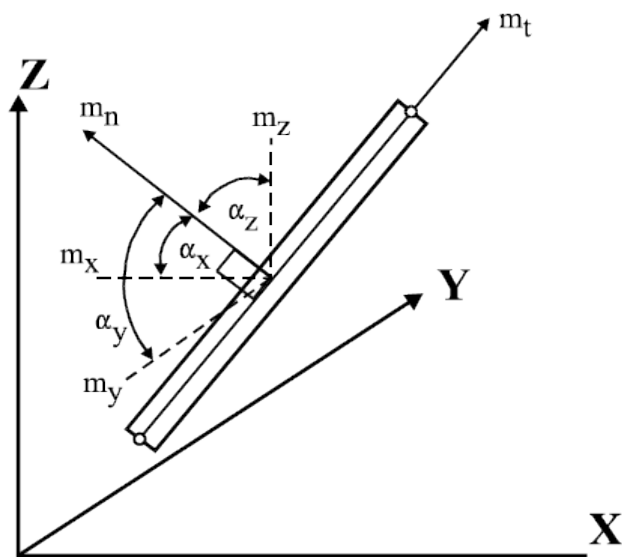


Figure 2. Beam element mass model

5 SAMPLE PROBLEMS

The structure shown in Figure 1 was used to illustrate various capabilities of the Dynpac program. Three separate Dynpac analyses are illustrated:

1. The dynamic characteristics of the structure submerged in water were determined using the consistent mass approach. Seastate override lines were used for the hydrodynamic modeling. The linearized foundation elements were included in the model file.
2. Sample Problem 2 is the same as Sample Problem 1 except that instead of modeling linearized pile stubs, a linearized foundation superelement was used. The ability to convert loads from any load case to mass without copying the load into LC 1 is also illustrated.
3. The natural modes of the deck in Figure 3 were determined using the lumped mass approach. Additional joint weight was added in the Dynpac input file.

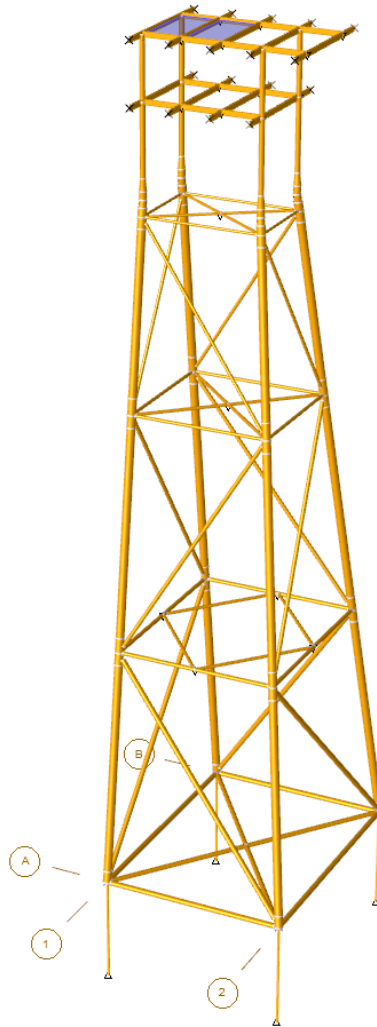


Figure 3. Jacket model with pile stubs

5.1 JACKET WITH PILE STUBS

The following example illustrates the use of the Seastate and Dynpac programs to determine the dynamic characteristics of a structure submerged in a fluid.

The structure in Figure 1 stands in 261.0 feet of salt water (density 64.2 lb/ft³). The member mass, mass of marine growth, mass of entrapped water and virtual or added mass were calculated automatically using the consistent mass approach. The Seastate program was used to determine the effective member properties including diameter, density, etc. to account for the hydrodynamic properties of the members.

Load cases consisting of equipment, area, and miscellaneous loads were specified in the SAC input file to account for unmodeled members and equipment weights that could affect the dynamic characteristics of the structure. Additional joint weights were specified in the Dynpac input file.

Dummy piles used to simulate the soil/pile interaction were developed using the Pile program and were added to the model. The degrees of freedom to be retained for determining the generalized masses and the eigenvectors were designated (using Precede) by specifying a '2' for the joint DOF.

A Seastate input file containing override lines to account for the hydrodynamics of unmodeled members and appurtenances was used as the SACS input file. 85% of load case EQPT and 75% of load case AREA in the model file were used to calculate the converted mass for the mode shape extraction.

The following is a portion of the Seastate input file:

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	LDOPT	NF+Z	64.30	490.05	-261.0	261.00	GLOB	DYN
3	* SELECTS LOAD CASES TO BE CONVERTED TO MASS							
4	LCSEL	DY	MISC EQPT AREA					
5	* DESIGNATES FACTOR ON LOAD CASES TO BE CONVERTED IF NOT 1.0							
6	LCFAC	DY	0.85	EQPT				
7	LCFAC	DY	0.75	AREA				
8	* USE LOADING IN MODEL FILE ONLY							
9	FILE J							
10	CDM	1.0	0.600	1.200	0.600	1.200		
11	CDM	100.0	0.600	1.200	0.600	1.200		
12	MGROV							
13	MGROV	0.00	200.000	1.000				
14	MGROV	200.0	261.000	2.000				
15	GRPOV							
16	GRPOVAL	LG1 F						
17	GRPOVAL	LG2 F						
18	GRPOVAL	LG3 F						
19	GRPOV	LG4 F						
20	GRPOV	PL1NF			0.001	0.001		
21	GRPOV	PL2NF			0.001	0.001		
22	GRPOV	PL3NF			0.001	0.001		
23	GRPOV	PL4NF			0.001	0.001		
24	GRPOV	W.BNF 0.001			0.001	0.001		
25	LOAD							
26	END							

The following is a description of the Seastate input file:

Line 1. The LDOPT line specifies the physical parameters of the structure such as water depth, water and steel density etc. 'DYN' in columns 56-58 specifies that a SACS hydrodynamic model is to be created for use by Dynpac.

Line 3. The LCSEL line designates that if the convert load case to mass option is specified in the Dynpac input file, only load cases MISC, EQPT, and AREA are to be converted.

Lines 5-6. The LCFAC line indicates that load case EQPT is to be factored by 0.85 and load case AREA is to be factored by 0.75 when converted to mass.

Line 8. The FILE line indicates that only loading in the jacket geometry file is to be considered for this analysis (i.e. 'J' in column 6).

Lines 9-24. The CDM, MGROV and GRPOV lines ensure that entrapped water mass and added or virtual mass are generated accurately.

The following is a portion of the model file used for this sample followed by a description of the input:

	1	2	3	4	5	6	7	8
1234567890123456789012345678901234567890123456789012345678901234567890								
1	SAMPLE	08	ENGLISH	UNITS	MODEL			
2	OPTIONS	EN	SDUC	2	1	PTPT	PT	
3	SECT							
4	SECT	CONE	CON			36.0000	75026.000	
5	SECT	PILSTUB	PRI165.1824786.0	24786.0	24786.0	10.0	10.0	
6	GRUP							
7	*****ADDITIONAL SACS GROUP LINES*****							
8	GRUP	STB	PILSTUB					
9	*****ADDITIONAL SACS GROUP LINES*****							
10	MEMBER							
11	MEMBER	2102S	102	STBSK				
12	MEMBER	OFFSETS				76.9		
13	MEMBER	2104S	104	STBSK				
14	MEMBER	OFFSETS				76.9		
15	MEMBER	2106S	106	STBSK				
16	MEMBER	OFFSETS				76.9		
17	MEMBER	2108S	108	STBSK				
18	MEMBER	OFFSETS				76.9		
19	*****ADDITIONAL SACS MEMBER LINES*****							
20	JOINT							
21	JOINT	101	-24.	-50.	-261.	-3.000		
22	JOINT	102	-24.	-50.	-261.			
23	JOINT	102S	-24.	-50.	-261.	-525.8	111111	
24	JOINT	103	51.	-50.	-261.	4.800	-3.000	
25	JOINT	104	51.	-50.	-261.	4.800	-3.000	
26	JOINT	104S	51.	-50.	-261.	4.800	-3.000	-525.8 111111
27	JOINT	105	-24.	50.	-261.	3.000		
28	JOINT	106	-24.	50.	-261.	3.000		
29	JOINT	106S	-24.	50.	-261.	3.000	-525.8 111111	
30	JOINT	107	51.	50.	-261.	4.800	3.000	
31	JOINT	108	51.	50.	-261.	4.800	3.000	
32	JOINT	108S	51.	50.	-261.	4.800	3.000	-525.8 111111
33	JOINT	109	13.	0.	-261.	8.400		222000
34	JOINT	201	-24.	-38.	-164.	-1.500		222000
35	JOINT	202	-24.	-38.	-164.	-1.500		
36	JOINT	203	41.	-38.	-164.	8.400	-1.500	222000
37	JOINT	204	41.	-38.	-164.	8.400	-1.500	
38	JOINT	205	-24.	38.	-164.	1.500		222000
39	JOINT	206	-24.	38.	-164.	1.500		
40	JOINT	207	41.	38.	-164.	8.400	1.500	222000
41	JOINT	208	41.	38.	-164.	8.400	1.500	
42	JOINT	209	-24.	0.	-164.			222000
43	JOINT	210	8.	38.	-164.	10.296	1.500	222000
44	JOINT	211	41.	0.	-164.	8.400		222000

```

45 JOINT 212      8.   -38.  -164. 10.296 -1.500      222000
46 JOINT 301    -24.   -26.   -69.   -3.000      222000
47 *****ADDITIONAL SACS JOINT LINES*****
48 LOAD
49 *****ADDITIONAL SACS LOAD LINES*****
50 LOADCNMISC
51 LOAD Z 712 703      -0.1900      -0.1900      GLOB UNIF  WALK1
52 LOAD Z 703 707      -0.1900      -0.1900      GLOB UNIF  WALK1
53 LOAD Z 707 723      -0.1900      -0.1900      GLOB UNIF  WALK1
54 LOAD Z 833 836      -0.1900      -0.1900      GLOB UNIF  WALK2
55 LOAD Z 836 839      -0.1900      -0.1900      GLOB UNIF  WALK2
56 LOAD Z 839 844      -0.1900      -0.1900      GLOB UNIF  WALK2
57 LOAD 807              -20.000              GLOB JOIN  CRANE
58 *
59 ***LDS1**      -8.000      20.000      50.000      -8.000      20.000      50.000
60 ***LDS2**              -10.000              34.000      0.100
61 ***LDS3**      0.100      1      2      2      0      0MISC  -1EQUPSKIDFIREWALLX
62 LOAD Z 705 720      3.95000-1.6667      GLOB CONC  FIREWALL
63 LOAD Z 705 720      4.05000-1.6667      GLOB CONC  FIREWALL
64 LOAD Z 717 721      3.95000-1.6667      GLOB CONC  FIREWALL
65 LOAD Z 717 721      4.05000-1.6667      GLOB CONC  FIREWALL
66 LOAD Z 718 722      3.95000-1.6667      GLOB CONC  FIREWALL
67 LOAD Z 718 722      4.05000-1.6667      GLOB CONC  FIREWALL
68 END

```

The following is a description of the SACS input file:

Line 5. The dummy pile section properties are defined using section PILSTUB.

Line 8. The dummy pile group PST is defined.

Lines 36-43. Dummy pile members 102S-102, 104S-104, 106S-106 and 108S-108 are defined.

Lines 23-32. The dummy pile bottom joints 102S, 104S, 106S and 108S are fixed (joint fixity '111111').

Lines 33-46. The retained degrees of freedom are specified by '2' in columns 55-60 on the appropriate JOINT lines. For example, Joint 109 is retained for translation in the X, Y, and Z directions as designated by '222000' in columns 55-60.

Lines 48-65. The loads of Load Condition 'MISC' account for the weight of unmodeled members and equipment and will be converted to masses by Dynpac.

Seastate and Dynpac were executed in succession to determine the dynamic characteristics of the structure. The output structural data file created by Seastate containing the effective member properties was used as the model input file for Dynpac.

Note: Seastate and Dynpac can be run as separate analysis steps or together as a single step. When executing separately, specify the Seastate output structural data file as the SACS input file for the Dynpac execution.

The following is the Dynpac input file used for this sample followed by a description of the input:

```

      1      2      3      4      5      6      7      8
1234567890123456789012345678901234567890123456789012345678901234567890
1 ENGLISH TEST MODEL
2 * SA-Z DESIGNATES Z LOADS TO BE CONVERTED TO MASS WITH OPPOSITE SIGN
3 DYNOPT +ZEN 15CONS490.0      -261.0 261.0      1.0      SA-Z
4 * WEIGHTS NOT DEFINED IN THE MODEL

```

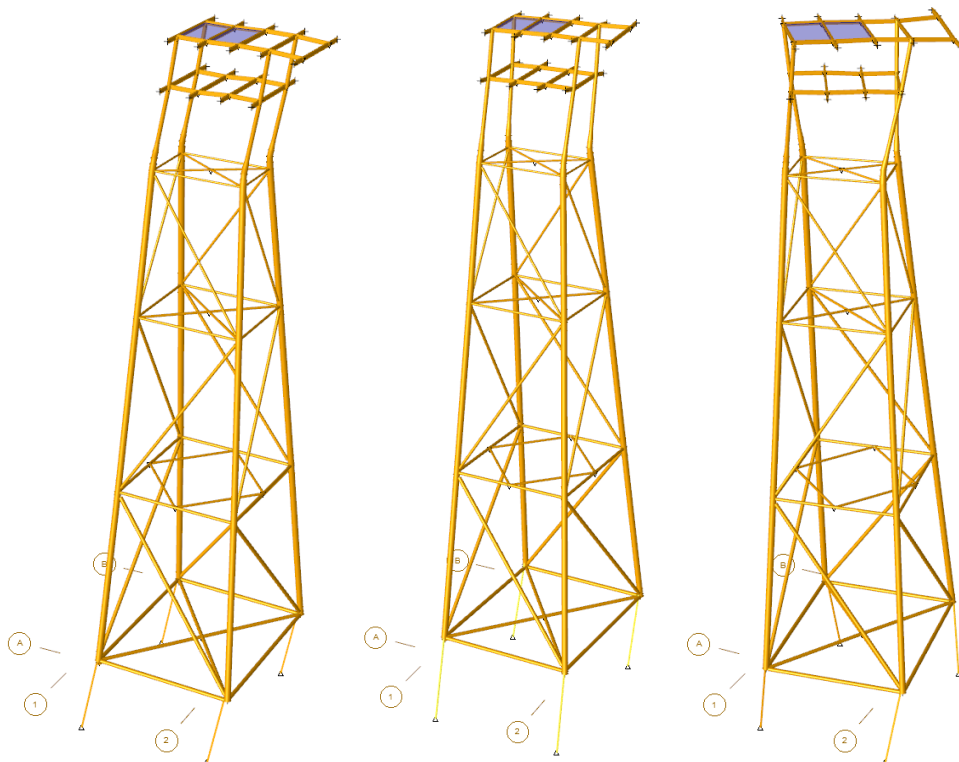
5	JTWGT	801	2.0	2.0	2.0
6	JTWGT	803	2.0	2.0	2.0
7	JTWGT	805	2.0	2.0	2.0
8	JTWGT	807	2.0	2.0	2.0
9	END				

Line 3. The DYNOPT line specifies the following:

- a. The vertical coordinate is the +Z direction and English units are to be used as specified in columns 8-9 and 10-11 respectively.
- b. 15 modes are desired (columns 12-14).
- c. The consistent mass approach is specified by 'CONS' in columns 15-18.
- d. The structure and fluid density are 490.0 and default (64.2 lb/ft³) respectively.
- e. The mudline elevation (-261.0) and the water depth (261.0) are specified in columns 33-39 and 40-46 respectively.
- g. Loads from the SACS data are to be used as masses and the Z direction masses will be opposite sign of the specified Z direction load ('SA-Z' in columns 63-66).

Lines 5-8. The JTWGT lines specify that four additional joint weights of 2 kips will be added to joints 801, 803, 805, and 807.

Six of the modes are displayed below. Excerpts from the output file follows.



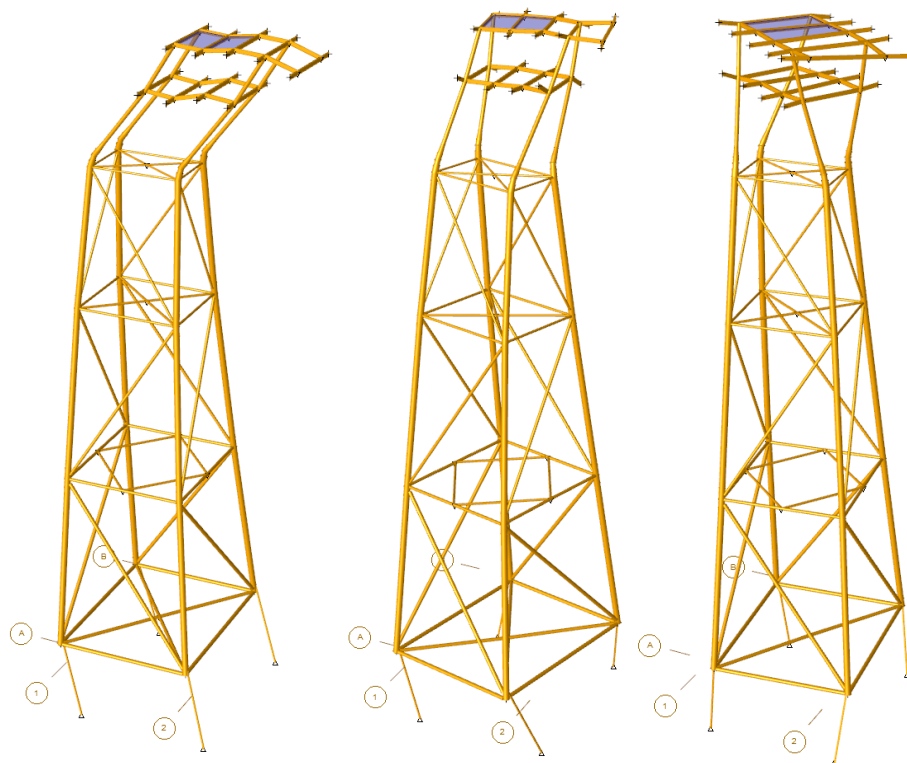


Figure 4. Jacket mode shapes

** DYNAMIC ANALYSIS PARAMETERS **

NO. JOINTS 87

NO. MEMBERS..... 153

NO. PLATES 2

NO. SHELL ELEMENTS..... 0

NO. REDUCED DOF..... 387

NO. RETAINED DOF..... 135

NO. NULL DOF..... 0

NO. MODES 15

NO. VECTORS..... 15

DYNPAC WEIGHT CONTINGENCY.. 1.000

SACS LOAD CONTINGENCY..... 1.000

ADDED WEIGHT CONTINGENCY... 1.000

EXPANDED MODE SHAPES REQUESTED

MASS PASSED FROM SACS DATA - ADDITIONAL

DIRECTION - -Z

AUTOMATIC CONSISTENT MASS OPTION SELECTED

STRUCTURAL DENSITY .. 490.000 LB/FT3

FLUID DENSITY 64.300 LB/FT3

ADDED MASS COEFFICIENTS

MEMBER NORMAL 1.000

MEMBER AXIAL 0.000

PLATE 1.000

MUDLINE ELEVATION = -261.00 FT

WATER DEPTH = 261.00 FT

ALL TUBULAR MEMBERS CONSIDERED NON-BUOYANT UNLESS OTHERWISE SPECIFIED

ENGLISH TEST MODEL

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SACS IV-FREQUENCIES AND GENERALIZED MASS

MODE	FREQ. (CPS)	GEN. MASS	EIGENVALUE	PERIOD (SECS)
1	0.309762	3.9727042E+03	2.6398845E-01	3.2282885
2	0.350484	3.6786831E+03	2.0620775E-01	2.8532010
3	0.568959	1.1307374E+03	7.8248940E-02	1.7575962
4	0.661552	2.8052165E+03	5.7877871E-02	1.5115975
5	0.666882	3.2945772E+03	5.6956442E-02	1.4995167
6	0.801626	1.1774440E+03	3.9418196E-02	1.2474647
7	1.351897	5.1110953E+03	1.3859681E-02	0.7397015
8	1.522383	3.3594363E+02	1.0929308E-02	0.6568651
9	1.535375	1.2470120E+02	1.0745127E-02	0.6513069
10	1.617849	2.3150009E+03	9.6775203E-03	0.6181045
11	1.980603	2.6553749E+02	6.4572143E-03	0.5048966
12	2.400655	1.4746010E+02	4.3952215E-03	0.4165530
13	2.476752	2.6696667E+02	4.1292872E-03	0.4037545
14	2.522800	2.4011208E+02	3.9799224E-03	0.3963850
15	2.555514	8.3777613E+02	3.8786789E-03	0.3913108

MASS PARTICIPATION FACTOR REPORT

BASED ON RETAINED DEGREES OF FREEDOM

MODE	***** MASS PARTICIPATION FACTORS *****			***** CUMULATIVE FACTORS *****		
	X	Y	Z	X	Y	Z
1	0.8639823	0.0000434	0.0006117	0.863982	0.000043	0.000612
2	0.0000318	0.8898029	0.0000157	0.864014	0.889846	0.000627
3	0.0002929	0.0073161	0.0000424	0.864307	0.897162	0.000670
4	0.1081960	0.0000329	0.0000025	0.972503	0.897195	0.000672
5	0.0002384	0.0854844	0.0001220	0.972741	0.982680	0.000794
6	0.0000062	0.0041994	0.0002404	0.972748	0.986879	0.001035
7	0.0267258	0.0000654	0.0165137	0.999473	0.986944	0.017548
8	0.0000249	0.0013498	0.0008927	0.999498	0.988294	0.018441
9	0.0000124	0.0000020	0.0629689	0.999511	0.988296	0.081410
10	0.0000824	0.0112332	0.0003442	0.999593	0.999529	0.081754
11	0.0000003	0.0001174	0.0000233	0.999593	0.999647	0.081777
12	0.0000020	0.0000001	0.0014304	0.999595	0.999647	0.083208
13	0.0000001	0.0000014	0.0005705	0.999595	0.999648	0.083778
14	0.0000002	0.0000012	0.0287099	0.999596	0.999650	0.112488
15	0.0000000	0.0000068	0.1209760	0.999596	0.999656	0.233464

5.2 JACKET WITH PILE SUPERELEMENT

The following example illustrates the ability to use an equivalent foundation super-element and to convert loading in any load case to mass, therefore eliminating the need to modify the model for dynamic analysis purposes. Only the degrees of freedom to be retained for determining the generalized masses and the eigen vectors were specified in the model by inputting a '2' in the appropriate joint fixity column.

Note: Because retaining DOFs has no effect on the model for static analysis, the same model file can be used for static and dynamic analyses.

The following is a portion of the model file to be sent through Seastate for hydrodynamic modeling. The differences between the model requirements for sample 1 and this sample are discussed below:

- A. Unlike Sample Problem 1, pile stub members 102S-102, 104S-104, 106S-106 and 108S-108 are not included in the model. An equivalent foundation super-element is to be used instead. Therefore, pile stub section 'PILSTUB' and pile stub group 'STB' are not required in the model input file.
- B. The foundation super-element input is specified in the Dynpac analysis runfile under the Solve options. This could be entered in the OPTIONS line of the model input file but using the analysis runfile options allows us to use the same model file for both the superelement creation and mode shape extraction analyses.
- C. The pile joints at the mudline are designated with PILEHD fixity.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	SAMPLE 08 ENGLISH UNITS MODEL							
3	OPTIONS	EN	SDUC	2 1	PTPT	PT		
4	SECT							
5	SECT CONE	CON			36.0000	75026.000		
6	GRUP							
7	*****ADDITIONAL SACS GROUP LINES*****							
8	GRUP W02 W24X131		29.0111	2035.97 1	1.001.00	0.500 490.00		
9	MEMBER							
10	MEMBER 101 201 LG1							
11	*****ADDITIONAL SACS MEMBER LINES*****							
12	JOINT							
13	JOINT 101	-24.	-50.	-261.	-3.000			
14	JOINT 102	-24.	-50.	-261.	-3.000	PILEHD		
15	JOINT 103	51.	-50.	-261.	4.800 -3.000			
16	JOINT 104	51.	-50.	-261.	4.800 -3.000	PILEHD		
17	JOINT 105	-24.	50.	-261.	3.000			
18	JOINT 106	-24.	50.	-261.	3.000	PILEHD		
19	JOINT 107	51.	50.	-261.	4.800 3.000			
20	JOINT 108	51.	50.	-261.	4.800 3.000	PILEHD		
21	JOINT 109	13.	0.	-261.	8.400	222000		
22	JOINT 201	-24.	-38.	-164.	-1.500	222000		
23	JOINT 202	-24.	-38.	-164.	-1.500			
24	JOINT 203	41.	-38.	-164.	8.400 -1.500	222000		
25	JOINT 204	41.	-38.	-164.	8.400 -1.500			
26	JOINT 205	-24.	38.	-164.	1.500	222000		
27	JOINT 206	-24.	38.	-164.	1.500			
28	JOINT 207	41.	38.	-164.	8.400 1.500	222000		
29	JOINT 208	41.	38.	-164.	8.400 1.500			
30	JOINT 209	-24.	0.	-164.		222000		
31	JOINT 210	8.	38.	-164.	10.296 1.500	222000		
32	JOINT 211	41.	0.	-164.	8.400	222000		

```

32 JOINT 212      8.   -38.  -164. 10.296 -1.500      222000
33 JOINT 301    -24.   -26.   -69.   -3.000      222000
34 *****ADDITIONAL SACS JOINT LINES*****
35 LOADCNMISC
36 LOAD Z 712 703      -0.1900      -0.1900      GLOB UNIF  WALK1
37 LOAD Z 703 707      -0.1900      -0.1900      GLOB UNIF  WALK1
38 LOAD Z 707 723      -0.1900      -0.1900      GLOB UNIF  WALK1
39 LOAD Z 833 836      -0.1900      -0.1900      GLOB UNIF  WALK2
40 LOAD Z 836 839      -0.1900      -0.1900      GLOB UNIF  WALK2
41 LOAD Z 839 844      -0.1900      -0.1900      GLOB UNIF  WALK2
42 LOAD 807              -20.000      GLOB JOIN  CRANE
43 *
44 ***LDS1**    -8.000    20.000    50.000    -8.000    20.000    50.000
45 ***LDS2**              -10.000              34.000    0.100
46 ***LDS3**    0.100    1    2    0    0MISC  -1EQUPSKIDFIREWALLX
47 LOAD Z 705 720    3.95000-1.6667      GLOB CONC  FIREWALL
48 LOAD Z 705 720    4.05000-1.6667      GLOB CONC  FIREWALL
49 LOAD Z 717 721    3.95000-1.6667      GLOB CONC  FIREWALL
50 LOAD Z 717 721    4.05000-1.6667      GLOB CONC  FIREWALL
51 LOAD Z 718 722    3.95000-1.6667      GLOB CONC  FIREWALL
52 LOAD Z 718 722    4.05000-1.6667      GLOB CONC  FIREWALL
53 END

```

As in Sample Problem 1, Seastate and Dynpac were executed in succession to determine the dynamic characteristics of the structure. The output structural data file created by Seastate containing the effective member properties was used as the SACS input file for Dynpac.

The following is the Dynpac input file followed by a detailed explanation of the features implemented:

```

1234567890123456789012345678901234567890123456789012345678901234567890
1 ENGLISH TEST MODEL
2 * SA-Z DESIGNATES Z LOADS TO BE CONVERTED TO MASS WITH OPPOSITE SIGN
3 DYNOPT +ZEN 15CONS490.0      -261.0 261.0    1.0      SA-Z
4 * WEIGHTS NOT DEFINED IN THE MODEL
5 JTWGT 801 2.0      2.0      2.0
6 JTWGT 803 2.0      2.0      2.0
7 JTWGT 805 2.0      2.0      2.0
8 JTWGT 807 2.0      2.0      2.0
9 END

```

Line 3. The DYNOPT line specifies the following:

- The vertical coordinate is the +Z direction and English units are to be used as specified in columns 8-9 and 10-11 respectively.
- 15 modes are desired (columns 12-14).
- The consistent mass approach is specified by 'CONS' in columns 15-18.
- The structure and fluid density are 490.0 and default (64.2 lb/ft³) respectively.
- The mudline elevation (-261.0) and the water depth (261.0) are specified in columns 33-39 and 40-46 respectively.
- Loads from the SACS data are to be used as masses and the Z direction masses will be opposite sign of the specified Z direction load ('SA-Z' in columns 63-66).

Lines 5-8. The JTWGT lines specify that four additional joint weights of 2 kips will be added to joints 801, 803, 805, and 807.

5.3 DECK STRUCTURE

The following example illustrates the use of the Dynpac program to determine the dynamic characteristics of a deck structure.

The deck of a structure modeled to the top of jacket elevation contains a piece of reciprocating machinery. The weight of the machinery was specified in Load Case MACH. Additional load cases were defined to represent the area loads, additional equipment and other unmodeled dead loads. The member mass and masses calculated from load cases will be applied as lumped masses. Figure 5 is a plot of the deck for this sample.

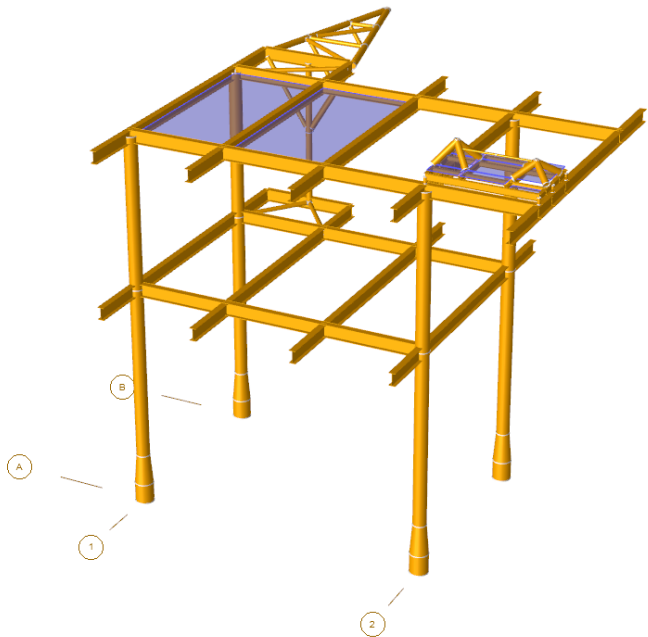


Figure 5. Deck model

The following are the steps required to execute the Dynpac analysis:

Miscellaneous loads to account for unmodeled members and equipment were will be converted to mass. The degrees of freedom to be retained for determining the generalized masses and the eigenvectors were designated (using Precede) by specifying a '2' for the joint DOF.

The following is the SACS input file used for the analysis:

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	SAMPLE	12	ENGLISH	UNITS	MODEL			
2	OPTIONS	EN		SDUC	2	1	PTPT	PT
3	LCSEL	DY		AREA	EQPT	MACH	MISC	
4	SECT							
5	SECT	CONE		CON			36.0000	.75026.000
6	GRUP							
7	GRUP	DUM		12.000	1.000	29.0011	.6036.00	9 1.001.00 0.500N490.00

The following is a description of the SACS input file:

Lines 32-35 The retained degrees of freedom are specified by '2' in columns 55-60 on the appropriate JOINT lines. For example, Joint 701 is retained for translation in the X, Y, and Z directions as designated by '222000' in columns 55-60.

Line 39. Load case MACH accounts for the equipment and will be converted to masses by Dynpac. The weight of the reciprocating machinery for example, was modeled as joint loads at joints 1000 and 3000.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	DYNOPT +ZEN 20CONS 490.		64.2				SA-Z	
2	END							

Line 1. The DYNOPT line specifies the following:

- a. The vertical coordinate is the +Z direction and English units are to be used as specified in columns 8-9 and 10-11 respectively.
- b. 20 modes are desired (columns 12-14).

c. The lumped mass approach is specified by 'CONS' in columns 15-18.

d. The structure density is 490.0 lb/ft³.

f. Loads from the SACS data are to be used as masses and the Z direction masses will be opposite sign of the specified Z direction load ('SA-Z' in columns 63-66).

Modes 1 through 4 are displayed below with excerpts of the Dynpac listing file following.

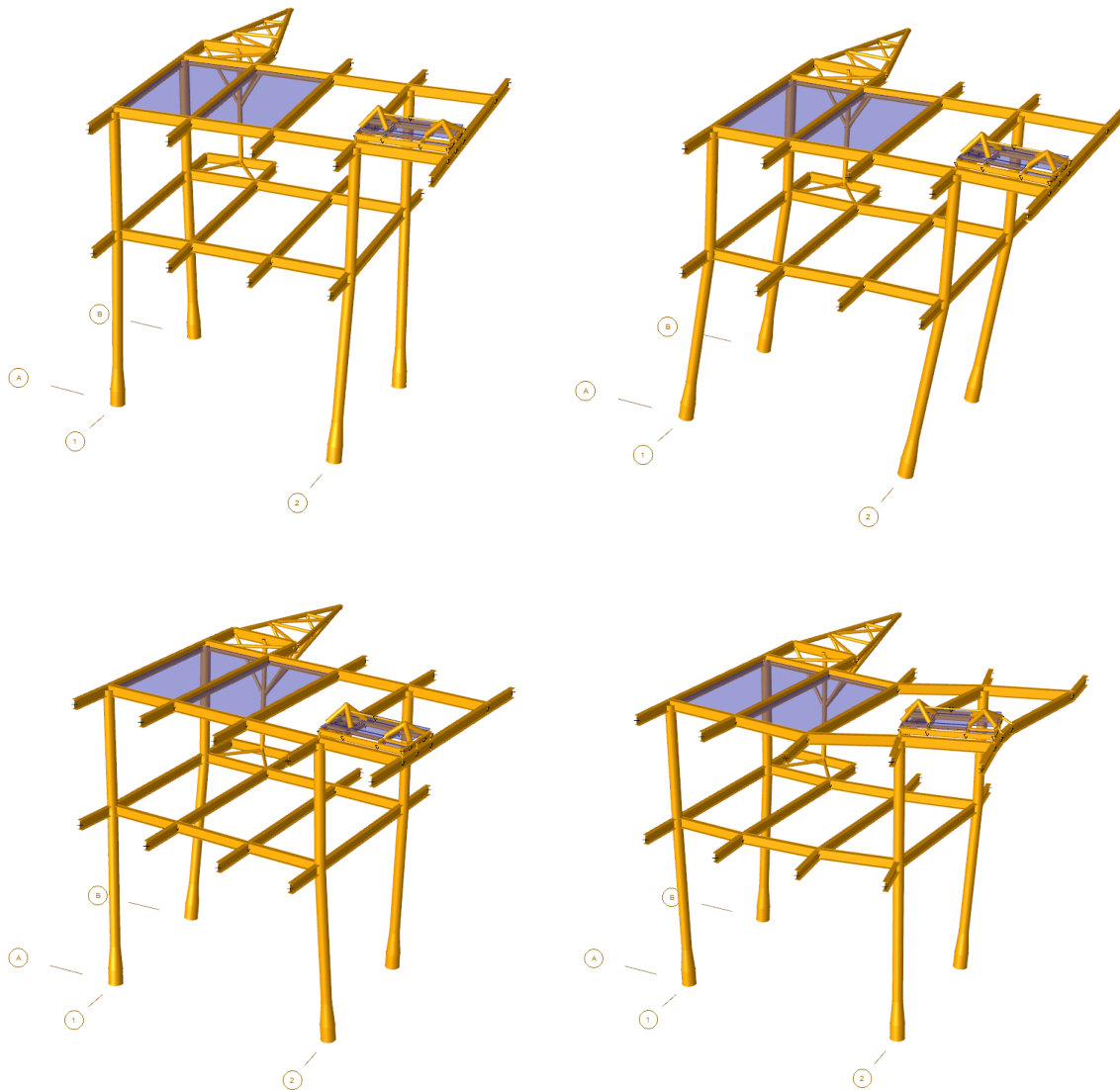


Figure 6. Deck mode shapes

** DYNAMIC ANALYSIS PARAMETERS **

NO. JOINTS 75
NO. MEMBERS..... 123
NO. PLATES 11
NO. SHELL ELEMENTS..... 0
NO. REDUCED DOF..... 276
NO. RETAINED DOF..... 174
NO. NULL DOF..... 0
NO. MODES 20
NO. VECTORS..... 20
DYNPAC WEIGHT CONTINGENCY.. 1.000
SACS LOAD CONTINGENCY..... 1.000
ADDED WEIGHT CONTINGENCY... 1.000

EXPANDED MODE SHAPES REQUESTED

MASS PASSED FROM SACS DATA - ADDITIONAL

DIRECTION - -Z

AUTOMATIC CONSISTENT MASS OPTION SELECTED

STRUCTURAL DENSITY .. 490.000 LB/FT3

SACS IV-FREQUENCIES AND GENERALIZED MASS

MODE	FREQ. (CPS)	GEN. MASS	EIGENVALUE	PERIOD (SECS)
1	0.325660	9.1779700E+02	2.3884171E-01	3.0706828
2	0.339435	1.8507796E+03	2.1985059E-01	2.9460743
3	0.406692	5.9514632E+02	1.5314710E-01	2.4588626
4	1.407973	5.3890858E+02	1.2777667E-02	0.7102409
5	1.882999	2.7977317E+02	7.1439745E-03	0.5310676
6	2.382503	2.8430248E+02	4.4624482E-03	0.4197266
7	2.463047	3.2707757E+02	4.1753680E-03	0.4060011
8	2.618373	2.4380048E+02	3.6946829E-03	0.3819165
9	2.755919	5.1319270E+02	3.3350878E-03	0.3628553
10	2.990387	6.1930932E+02	2.8326011E-03	0.3344049
11	3.355452	2.8320782E+02	2.2497703E-03	0.2980224
12	3.551919	1.0954029E+02	2.0077704E-03	0.2815379
13	3.877866	2.9913966E+02	1.6844364E-03	0.2578738
14	4.353015	8.3178932E+01	1.3367800E-03	0.2297258
15	4.439922	1.0944511E+02	1.2849596E-03	0.2252292
16	4.864130	2.8356462E+01	1.0706068E-03	0.2055866
17	5.281648	1.2013413E+02	9.0803225E-04	0.1893348
18	5.765977	4.0384968E+01	7.6189390E-04	0.1734312
19	6.142706	1.6725919E+02	6.7130634E-04	0.1627947
20	6.418559	2.3899120E+02	6.1484437E-04	0.1557982

MASS PARTICIPATION FACTOR REPORT

BASED ON RETAINED DEGREES OF FREEDOM

MODE	***** MASS PARTICIPATION FACTORS *****			***** CUMULATIVE FACTORS *****		
	X	Y	Z	X	Y	Z
1	0.0389276	0.7559233	0.0000002	0.038928	0.755923	0.000000
2	0.9535053	0.0342457	0.0000641	0.992433	0.790169	0.000064
3	0.0004347	0.2034694	0.0000118	0.992868	0.993638	0.000076
4	0.0000922	0.0004339	0.0000047	0.992960	0.994072	0.000081
5	0.0000700	0.0037727	0.0000223	0.993030	0.997845	0.000103
6	0.0018269	0.0001662	0.0000092	0.994857	0.998011	0.000112
7	0.0000864	0.0000404	0.0004467	0.994943	0.998052	0.000559
8	0.0004744	0.0000195	0.0588480	0.995418	0.998071	0.059407
9	0.0031329	0.0000018	0.0605284	0.998550	0.998073	0.119935
10	0.0009347	0.0000037	0.0428080	0.999485	0.998077	0.162743
11	0.0000745	0.0000837	0.0230568	0.999560	0.998160	0.185800
12	0.0000447	0.0004144	0.0122426	0.999604	0.998575	0.198043
13	0.0000636	0.0000482	0.1113421	0.999668	0.998623	0.309385
14	0.0002765	0.0000224	0.1331073	0.999944	0.998645	0.442492
15	0.0000023	0.0008281	0.0096637	0.999947	0.999473	0.452156
16	0.0000007	0.0004011	0.0382295	0.999947	0.999874	0.490386
17	0.0000150	0.0000020	0.0920545	0.999962	0.999876	0.582440
18	0.0000000	0.0000035	0.0123412	0.999962	0.999880	0.594781
19	0.0000004	0.0000595	0.0203125	0.999963	0.999940	0.615094
20	0.0000004	0.0000592	0.0299106	0.999963	0.999999	0.645004

6 INPUT LINES

ADDITIONAL DYNAMIC MODAL EXTRACTION OPTIONS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL IN ANY DYNAMIC CHARACTERISTICS EXECUTION. IT SPECIFIES ADDITIONAL OVERALL ANALYSIS OPTIONS AND PARAMETERS.
(8-13)	ENTER THE DYNPAC WEIGHT CONTINGENCY FACTOR FOR USE IN THIS ANALYSIS. THIS FACTOR WILL BE USED TO MULTIPLY THE STRUCTURAL MASS GENERATED BY DYNPAC INCLUDING ANY DYNPAC ADDED WEIGHTS TO ACCOUNT FOR ANY CONTINGENCIES.
(14-19)	ENTER THE LOAD WEIGHT CONTINGENCY FACTOR FOR USE IN THIS ANALYSIS. THIS FACTOR WILL BE USED TO MULTIPLY THE MASSES FROM SACS IV LOADS TO ACCOUNT FOR ANY CONTINGENCIES.
(20-25)	ENTER THE INCLUDED WEIGHT CONTINGENCY FACTOR FOR USE IN THIS ANALYSIS. THIS FACTOR WILL BE USED TO MULTIPLY THE MASSES FROM SACS IV INCLUDED WEIGHTS TO ACCOUNT FOR ANY CONTINGENCIES.
NOTE:	THE LOAD CASE FACTOR IS APPLIED IN ADDITION TO ANY LOAD CASE FACTOR SPECIFIED BY 'LCFAC' LINES.
(74-76)	ENTER 'MST' TO INCLUDE MEMBER STRUCTURAL TORSIONAL MASS IN. MODEL MASS MATRIX.
(78-80)	ENTER 'DSE' TO GENERATE THE DATA FILE FOR DYNAMIC SUPERELEMENT.

LINE LABEL	WEIGHT CONTINGENCY FACTORS			LEAVE BLANK	MEMBER STRUCTURAL TORSIONAL MASS	DYNAMIC SUPERELEMENT
	DYNPAC CALCULATED STRUCTURAL MASSES	SACS IV LOAD MASSES	SACS IV INCLUDED WEIGHT MASSES			
DYNOP2						
1-- 6	8<--13	14<--19	20<--25	26-----73	74-76	78--80
DEFAULT	1	1	1			

DYNAMIC MODAL EXTRACTION OPTIONS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN ANY DYNAMIC CHARACTERISTICS EXECUTION. IT SPECIFIES THE OVERALL ANALYSIS OPTIONS AND PARAMETERS.
(8- 9)	ENTER THE VERTICAL COORDINATE DIRECTION SUCH AS +Z FOR THE Z COORDINATE IN THE POSITIVE VERTICAL DIRECTION.
(10-11)	ENTER THE UNITS DESIRED FOR THIS ANALYSIS. IF LEFT BLANK, THE PROGRAM WILL USE THE UNITS SPECIFIED ON THE SACS IV 'OPTIONS' LINE. 'EN' - ENGLISH UNITS (FEET, LBS, ETC.) 'MN' - METRIC UNITS (METERS, KILONEWTONS, ETC.) 'MS' - METRIC UNITS (METERS, KILOGRAMS, ETC.)
(12-14)	ENTER THE NUMBER OF MODES DESIRED.
(15-18)	ENTER THE MASS CALCULATION OPTION. 'LUMP' - LUMPED MASS (DIAGONAL MASS MATRIX) 'CONS' - CONSISTENT MASS NOTE THAT CONSISTENT MASS IS HIGHLY DESIRABLE FOR STRUCTURES THAT ARE IMMERSED IN A FLUID.
(19-25)	ENTER THE DEFAULT STRUCTURAL DENSITY FOR ALL PLATE AND SHELL ELEMENTS AND ANY BEAM ELEMENTS THAT DO NOT HAVE DENSITY SPECIFIED ON THE 'MEMBER' OR 'GRUP' LINE. THE DENSITY OF INDIVIDUAL BEAM, PLATE OR SHELL ELEMENTS CAN BE OVERRIDDEN WITH SUBSEQUENT OVERRIDE LINES.
(26-32)	IF THE STRUCTURE IS IN A FLUID, ENTER THE FLUID DENSITY IF NOT PREVIOUSLY SPECIFIED OR IF THE SPECIFIED VALUE IS TO BE OVERRIDDEN.
(33-39)	IF THE STRUCTURE IS IN A FLUID, ENTER THE MUDLINE ELEVATION IF NOT PREVIOUSLY SPECIFIED OR IF THE SPECIFIED VALUE IS TO BE OVERRIDDEN. THIS IS THE STRUCTURAL COORDINATE VALUE IN THE POSITIVE VERTICAL DIRECTION.
(40-46)	IF THE STRUCTURE IS IMMERSED IN FLUID, ENTER THE WATER DEPTH IF NOT PREVIOUSLY SPECIFIED OR IF THE VALUE IS TO BE OVERRIDDEN.

COLUMNS	COMMENTS
(47-48)	ENTER 'FL' OR 'NF' TO CONSIDER MEMBERS FLOODED OR NON-FLOODED IF IT IS NOT ALREADY SPECIFIED ON THE MEMBER GROUP INPUT LINE OR MEMBER INPUT LINE IN THE SACS MODEL FILE. LEAVING BLANK IS EQUIVALENT TO 'FL'. NOTE: THIS OPTION HAS NO EFFECT IF THE SEASTATE RUNS TO CREATE AN OCI FILE BEFORE RUNNING DYNPAC.
(49-53)	ENTER THE ADDED MASS COEFFICIENT FOR MEMBERS SURROUNDED BY FLUID. THIS VALUE IS FOR THE MOTION OF THE STRUCTURE AND IS NOT NORMALLY THE SAME AS IF THE STRUCTURE IS IMMERSSED IN AN ACCELERATING FLUID.
(54-58)	ENTER THE AXIAL ADDED MASS COEFFICIENT FOR MEMBERS SURROUNDED BY FLUID. THIS VALUE IS FOR THE AXIAL MOTION OF THE STRUCTURE. THE DEFAULTS ARE 0.0 FOR CONSISTENT MASS AND 1.0 FOR LUMPED MASS OPTIONS.
(59-62)	ENTER THE DEFAULT ADDED MASS COEFFICIENT FOR PLATES SURROUNDED BY FLUID. IF LEFT BLANK, THE VALUE INPUT IN COLUMNS 49-53 IS USED.
(63-68)	ENTER THE EXECUTION OPTIONS DESIRED: ' ' - USE ONLY MASSES CALCULATED BY DYNPAC. 'SA' - USE LOADS IN THE SACS IV DATA AS MASSES IN ADDITION TO MASSES CALCULATED BY DYNPAC. 'SO' - USE LOADS IN THE SACS IV DATA AS MASSES AND DO NOT USE ANY MASSES CALCULATED BY DYNPAC. '+X' - THE LOADS TO BE USED FROM THE SACS IV DATA FOR MASSES ARE THE X-DIRECTION LOADS ONLY, SAME SIGN AS LOAD RECORD. USE '+Y' AND '+Z' SIMILARLY. '-X' - THE LOADS TO BE USED FROM THE SACS IV DATA FOR MASSES ARE THE X-DIRECTION LOADS ONLY, OPPOSITE SIGN FROM LOAD RECORD. USE '-Y' AND '-Z' SIMILARLY.
(69-80)	ENTER THE OUTPUT OPTIONS DESIRED: 'MA' - PRINT MASS MATRIX. 'ST' - PRINT STIFFNESS MATRIX. 'OT' - PRINT ORTHOGONALITY CHECK MATRIX. 'RM' - CREATE REACTION MATRIX OUTPUT. THIS IS USED FOR TRANSFER FUNCTION CALCULATION IN DYNAMIC RESPONSE. THESE OPTIONS CAN BE ENTERED IN ANY ORDER.

[illegible]

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

MEMBER GROUP MASS OVERRIDE

COLUMNS	COMMENTARY
LOCATION	THIS INPUT IS OPTIONAL AND SHOULD FOLLOW THE PLOVR DATA IF IT IS INPUT. IF IT IS NOT INPUT, THIS INPUT SHOULD FOLLOW THE DYNOPT LINE.
GENERAL	THIS INPUT IS OPTIONAL. NORMALLY SEASTATE HAS BEEN EXECUTED BEFORE DYNPAC SO THAT ANY FLOODING STATUS, DENSITY DATA AND EFFECTIVE DIAMETER FOR FLUID ADDED MASS CALCULATIONS ARE INCLUDED ON THE MEMBER LINES IN THE SACS IV MODEL OUTPUT BY SEASTATE. DYNPAC USES THIS INFORMATION TO GENERATE MASS DATA. THIS INPUT LINE ENABLES THE USER TO SPECIFY THE WEIGHT DENSITY, BUOYANCY AND/OR OUTSIDE DIAMETER TO BE DIFFERENT THAN THAT PREVIOUSLY DESIGNATED FOR A PARTICULAR GROUP OF MEMBERS. THE DATA SPECIFIED HERE WILL COMPLETELY REPLACE ALL PREVIOUS INPUT DATA INCLUDING SPECIFIED DEFAULTS. THIS INPUT LINE SET CONSISTS OF GROUP MASS DATA OVERRIDE LINES FOR EACH GROUP WHOSE DATA IS BEING REPLACED.
(7- 9)	ENTER GROUP IDENTIFIER CODE FOR MEMBER GROUP WHOSE DENSITY, BUOYANCY AND/OR OUTSIDE DIAMETER ARE TO BE OVERRIDDEN. ANY GROUP IDENTIFIER CODE ENTERED HERE MUST CORRESPOND TO A GROUP CODE IN THE SACS IV MODEL FILE.
(11)	ENTER 'N' TO INDICATE THIS GROUP OF MEMBERS IS NOT FLOODED. AN 'F' IN THESE COLUMNS WILL INDICATE THAT MEMBERS WITH THIS GROUP IDENTIFIER ARE FLOODED. IF LEFT BLANK, IT WILL NOT CHANGE THE FLOODING ON THESE MEMBERS.
(13-20)	ENTER HERE THE OUTSIDE DIAMETER FOR THE SPECIFIED MEMBER GROUP THAT IS TO BE USED IN CALCULATING FLUID ADDED MASS. IF LEFT BLANK THE DEFAULT IS THE STRUCTURAL O.D. FOR TUBULARS AND ZERO FOR PRISMATICS OR THE FLUID ADDED MASS O.D., IF SPECIFIED ON THE MEMBER LINES. THIS INPUT HAS NO EFFECT ON THE STRUCTURAL PROPERTIES OF THE MEMBERS.
(21-30)	ENTER STRUCTURAL DENSITY IN WEIGHT PER UNIT VOLUME. IF LEFT BLANK OR ZERO, THE DENSITY WILL BE UNCHANGED.

LINE LABEL	GROUP CODE	FLOODED CRITERIA 'F' - FLOODED 'N' - NON-FLOODED	OUTSIDE DIAMETER FOR FLUID ADDED MASS CALCULATION	STRUCTURAL WEIGHT DENSITY	LEAVE BLANK
GROVR					
1-- 5	7-- 9	11	13--20	21--30	31-----80
DEFAULT			SEE NOTE FOR COLUMNS 13-20		
ENGLISH			IN	LB/CU.FT	
METRIC			CM	TONNE/CU.M	

JOINT CONCENTRATED WEIGHT

COLUMNS	COMMENTARY
GENERAL	THIS OPTIONAL INPUT LINE ENABLES THE USER TO SPECIFY ADDITIONAL WEIGHT AND WEIGHT MOMENT OF INERTIA DATA TO BE INPUT AS LUMPED DATA FOR SPECIFIED DEGREES OF FREEDOM. THIS DATA IS ADDED TO THE SYSTEM DATA GENERATED BY THE PROGRAM. A SEPARATE LINE SHOULD BE CODED FOR EACH JOINT TO WHICH MASS IS ADDED. THIS LUMPED DATA MAY BE PLACED ON EITHER RETAINED OR REDUCED FREE DEGREES OF FREEDOM.
(7-10)	ENTER THE JOINT NAME. THIS JOINT NAME SHOULD CORRESPOND TO A JOINT NAME FOUND IN THE SACS IV MODEL. RIGHT JUSTIFY INPUT.
(11-40)	ENTER WEIGHTS FOR 'X','Y' AND 'Z' DIRECTION TRANSLATIONAL DEGREES OF FREEDOM, RESPECTIVELY, FOR THIS JOINT.
(41-70)	ENTER WEIGHT MOMENTS OF INERTIA FOR 'X','Y' AND 'Z' ROTATIONAL DEGREES OF FREEDOM, RESPECTIVELY, FOR THIS JOINT.
(72-75)	ENTER JOINT A OF A MEMBER TO DEFINE JOINT WEIGHT IN THE LOCAL COORDINATES OF A MEMBER. LEAVE BLANK TO DEFINE JOINT WEIGHT IN GLOBAL COORDINATES.
(77-80)	ENTER JOINT B OF A MEMBER TO DEFINE JOINT WEIGHT IN THE LOCAL COORDINATES OF A MEMBER. LEAVE BLANK TO DEFINE JOINT WEIGHT IN GLOBAL COORDINATES.

LINE LABEL	JOINT NAME	JOINT CONCENTRATED WEIGHT DATA						MEMBER FOR LOCAL AXIS	
		WEIGHT (X DIRECTION)	WEIGHT (Y DIRECTION)	WEIGHT (Z DIRECTION)	WEIGHT MOMENT OF INERTIA (X AXIS)	WEIGHT MOMENT OF INERTIA (Y AXIS)	WEIGHT MOMENT OF INERTIA (Z AXIS)	JOINT A	JOINT B
JTWGT									
1-- 5	7-->10	11<--20	21<--30	31<--40	41<--50	51<--60	61<--70	72<--75	77<--80
DEFAULT		0	0	0	0	0	0		
ENGLISH		KIP	KIP	KIP	KIP-IN**2	KIP-IN**2	KIP-IN**2		
METRIC		TONNE	TONNE	TONNE	TONNE-CM**2	TONNE-CM**2	TONNE-CM**2		

MEMBER MASS OVERRIDE

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS OPTIONAL. NORMALLY SEASTATE HAS BEEN EXECUTED BEFORE DYNPAC SO THAT ANY FLOODING STATUS, DENSITY DATA AND EFFECTIVE DIAMETER FOR FLUID ADDED MASS CALCULATIONS ARE INCLUDED ON THE MEMBER LINES IN THE SACS IV MODEL OUTPUT BY SEASTATE. DYNPAC USES THIS INFORMATION TO GENERATE MASS DATA. THIS INPUT DATA ENABLES THE USER TO SPECIFY THE WEIGHT DENSITY, BUOYANCY AND OR OUTSIDE DIAMETER TO BE DIFFERENT THAN THAT PREVIOUSLY DESIGNATED FOR PARTICULAR MEMBERS. THIS INPUT CONSISTS OF MEMBER MASS DATA OVERRIDE LINES FOR EACH MEMBER WHOSE DATA IS BEING REPLACED. (7) ENTER 'N' TO INDICATE THAT THIS MEMBER IS NOT FLOODED. AN 'F' IN THESE COLUMNS WILL INDICATE THAT THIS MEMBER IS FLOODED. THE BUOYANT OR FLOODED CONDITION SPECIFIED ON THIS LINE FOR THIS LINE WILL OVERRIDE ANY CONDITIONS SPECIFIED IN THE 'DYNOPT' OR 'GROVR' LINES. IF LEFT BLANK, THE MEMBER FLOODED CONDITION WILL REMAIN UNCHANGED. (8-11) ENTER MEMBER START JOINT. THIS JOINT SHOULD CORRESPOND TO THE MEMBER START JOINT IN SACS IV DATA FILE. (12-15) ENTER MEMBER END JOINT. THIS JOINT SHOULD CORRESPOND TO THE MEMBER END JOINT IN SACS IV DATA FILE. (16-21) ENTER HERE THE OUTSIDE DIAMETER FOR THE SPECIFIED MEMBER THAT IS TO BE USED IN CALCULATING FLUID ADDED MASS. IF LEFT BLANK THE DEFAULT IS THE STRUCTURAL O.D. FOR TUBULARS AND ZERO FOR PRISMATICS OR THE FLUID ADDED MASS O.D., IF SPECIFIED ON THE MEMBER LINES. THIS INPUT HAS NO EFFECT ON THE STRUCTURAL PROPERTIES OF THE MEMBERS. (22-30) ENTER STRUCTURAL DENSITY IN WEIGHT PER UNIT VOLUME. IF LEFT BLANK OR ZERO, THE DENSITY WILL REMAIN UNCHANGED.

LINE LABEL	FLOODED CRITERIA 'F' - FLOODED 'N' - NON-FLOODED	*2MEMBER DESIGNATION		OUTSIDE DIAMETER FOR FLUID ADDED MASS CALCULATION	STRUCTURAL WEIGHT DENSITY	LEAVE BLANK
		START JOINT	END JOINT			
MBOVR						
1-- 5	7	8-->11	12-->15	16<--21	22<--30	45----80
DEFAULT				SEE NOTE FOR COLUMNS 16-21		
ENGLISH				IN	LB/CU.FT	
METRIC				CM	TONNE/CU.M	

PLATE GROUP OVERRIDE DATA

COLUMNS	COMMENTARY
LOCATION	THIS INPUT DATA IS OPTIONAL AND SHOULD FOLLOW THE DYNOPT INPUT LINE.
GENERAL	THIS OPTIONAL INPUT ENABLES THE USER TO SPECIFY FOR ANY PLATE GROUP PROPERTIES THAT ARE DIFFERENT FROM THOSE SPECIFIED ON THE 'DYNOPT' LINE.
(7- 9)	ENTER THE PLATE GROUP IDENTIFIER OF THOSE PLATES WHOSE PROPERTIES ARE BEING SPECIFIED. THIS IDENTIFIER MUST CORRESPOND TO A PLATE GROUP IDENTIFIER IN THE SACS IV DATA FILE.
(21-30)	ENTER STRUCTURAL WEIGHT DENSITY. IF LEFT BLANK OR ZERO, THE DEFAULT DENSITY IS THE DENSITY ENTERED ON THE 'DYNOPT' LINE.
NOTE:	THE DENSITY SPECIFIED ON THE 'DYNOPT' LINE IS USED AS THE DEFAULT DENSITY FOR PLATE ELEMENTS.
(31-35)	ENTER THE ADDED MASS COEFFICIENT IF DIFFERENT FROM THAT ENTERED ON THE 'DYNOPT' LINE. NOTE THAT A ZERO OR BLANK WILL RESULT IN NO CHANGE.

LINE LABEL	PLATE GROUP NAME	STRUCTURAL WEIGHT DENSITY	ADDED MASS COEFF.	LEAVE BLANK
PGOVR				
1-- 5	7-- 9	21<--30	31<--35	36-----80
DEFAULT		DYNOPT'		
ENGLISH		LB/CU.FT		
METRIC		TONNE/CU.M		

PLATE OVERRIDE DATA

COLUMNS	COMMENTARY
LOCATION	THIS INPUT DATA IS OPTIONAL AND SHOULD FOLLOW THE DYNOPT INPUT LINE.
GENERAL	THIS OPTIONAL INPUT ENABLES THE USER TO SPECIFY FOR ANY SET OF PLATES PROPERTIES THAT ARE DIFFERENT FROM THOSE SPECIFIED ON THE DYNOPT LINE.
(7-10)	ENTER NAME OF FIRST PLATE IN RANGE OF PLATES BEING SPECIFIED. THIS NAME MUST CORRESPOND TO A PLATE NAME IN THE SACS IV DATA FILE. IF THIS DENSITY APPLIES TO ALL PLATES, THEN ENTER '*****' IN COLUMNS 7-10 AND LEAVE COLUMNS 11-14 BLANK.
(11-14)	ENTER NAME OF LAST PLATE IN RANGE OF PLATES BEING SPECIFIED. THIS NAME MUST CORRESPOND TO A PLATE NAME IN THE SACS IV MODEL. ALL PLATES IN THE MODEL WHICH PHYSICALLY LIE BETWEEN THE FIRST AND LAST SPECIFIED PLATES WILL BE INCLUDED IN THIS PLATE RANGE. IF THIS FIELD IS LEFT BLANK, THE SINGLE PLATE SPECIFIED IN COLUMNS 7-10 WILL BE USED.
(21-30)	ENTER THE STRUCTURAL WEIGHT DENSITY. IF LEFT BLANK OR ZERO, THE DEFAULT DENSITY IS THE DENSITY ENTERED ON THE DYNOPT LINE.
NOTE:	THE DENSITY SPECIFIED ON THE DYNOPT LINE IS USED AS THE DEFAULT DENSITY FOR PLATE ELEMENTS.
(31-35)	ENTER THE ADDED MASS COEFFICIENT IF DIFFERENT FROM THAT ENTERED ON THE DYNOPT LINE. NOTE THAT A ZERO OR BLANK WILL RESULT IN NO CHANGE.

LINE LABEL	PLATE RANGE		STRUCTURAL WEIGHT DENSITY	ADDED MASS COEFF.	LEAVE BLANK
	START NAME	END NAME			
PLOVR					
1-- 5	7--10	11--14	21<--30	31<--35	36-----80
DEFAULT			DYNOPT'		
ENGLISH			LB/CU.FT		
METRIC			TONNE/CU.M		

SHELL MASS DENSITY

COLUMNS	COMMENTARY
GENERAL	THIS INPUT IS OPTIONAL INPUT WHICH ENABLES THE USER TO SPECIFY FOR ANY SET OF SHELLS WEIGHT DENSITIES THAT ARE DIFFERENT FROM THE SYSTEM WEIGHT DENSITY. AN INPUT WEIGHT DENSITY ON THIS LINE WILL OVERRIDE THE SYSTEM WEIGHT DENSITY FOR THE SPECIFIED RANGE OF SHELLS.
(7-10)	ENTER NAME OF FIRST SHELL IN RANGE OF SHELLS WHOSE DENSITY IS BEING SPECIFIED. THIS NAME MUST CORRESPOND TO A SHELL NAME IN THE SACS IV DATA FILE. IF THIS DENSITY APPLIES TO ALL SHELLS, THEN ENTER '*****' IN COLUMNS 7-10 AND LEAVE COLUMNS 11-14 BLANK.
(11-14)	ENTER NAME OF LAST SHELL IN RANGE OF SHELLS WHOSE DENSITY IS BEING SPECIFIED. THIS NAME MUST CORRESPOND TO A SHELL NAME IN THE SACS IV DATA FILE. ALL SHELLS IN THE FILE WHICH PHYSICALLY LIE BETWEEN THE FIRST AND LAST SPECIFIED SHELLS WILL BE INCLUDED IN THIS SHELL RANGE. IF THIS FIELD IS LEFT BLANK, THE SINGLE SHELL SPECIFIED IN COLUMNS 7-10 WILL BE USED.
(21-30)	ENTER THE STRUCTURAL WEIGHT DENSITY. IF LEFT BLANK, THE DEFAULT DENSITY IS THE DENSITY ENTERED ON THE DYNOPT LINE.
NOTE:	THE DENSITY SPECIFIED ON THE DYNOPT LINE IS USED AS THE DEFAULT DENSITY FOR SHELL ELEMENTS. THE DENSITY SPECIFIED ON THE SHELL LINE IN THE MODEL FILE IS NOT USED FOR THE PURPOSE OF DETERMINING SHELL MASS.

LINE LABEL	SHELL RANGE		STRUCTURAL WEIGHT DENSITY	LEAVE BLANK
	START NAME	END NAME		
SHOVR				
1-- 5	7--10	11--14	21<--30	31-----80
DEFAULT			DYNOPT'	
ENGLISH			LB/CU.FT	
METRIC			TONNE/CU.M	

DYNPAC TITLE

<u>COLUMNS</u>	<u>COMMENTARY</u>
LOCATION	IF INPUT, THIS OPTIONAL LINE IS FIRST IN THE DYNPAC INPUT FILE.
GENERAL	THIS LINE IS OPTIONAL AND ALLOWS THE USER TO SPECIFY A TITLE FOR DYNPAC OUTPUT OTHER THAN THE TITLE FROM THE SACS IV FILE.
(2-80)	ENTER ANY ALPHANUMERIC TITLE. THIS TITLE WILL APPEAR ON ALL PAGES OF DYNPAC OUTPUT.

DESCRIPTIVE TITLE

2-----80

WINDMILL EXECUTION OPTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO CREATE 6X6 MASS, STIFFNESS, AND DAMPING MATRICES FOR WINDMILL ANALYSIS. THE NUMBER OF RETAINED DEGREES OF FREEDOM MUST BE 6 AND FROM ONE JOINT.
(8-13)	ENTER THE DAMPING RATIO (PERCENT CRITICAL) FOR THE DAMPING MATRIX.
(14-15)	ENTER THE UNITS FOR THE OUTPUT MATRICES. LEAVE BLANK FOR OPTION ENTERED ON DYNOPT LINE. 'EN' - ENGLISH UNITS (INCHES, LBS, SLINCHES.) 'MN' - METRIC UNITS (METERS, NEWTONS, KILOGRAMS) 'MS' - METRIC UNITS (METERS, NEWTONS, KILOGRAMS)
(16-18)	ENTER THE DESIRED OUTPUT FORMAT. 'STD' - SACS STANDARD 'EU1' - EUROPEAN NO. 1

LINE LABEL	DAMPING	UNITS	OUTPUT FORMAT OPTION	LEAVE BLANK
WMILL				
1-- 5	8--13	14--15	16--18	19-----80
DEFAULT	1	DYNOPT	STD	
English	%			
Metric	%			